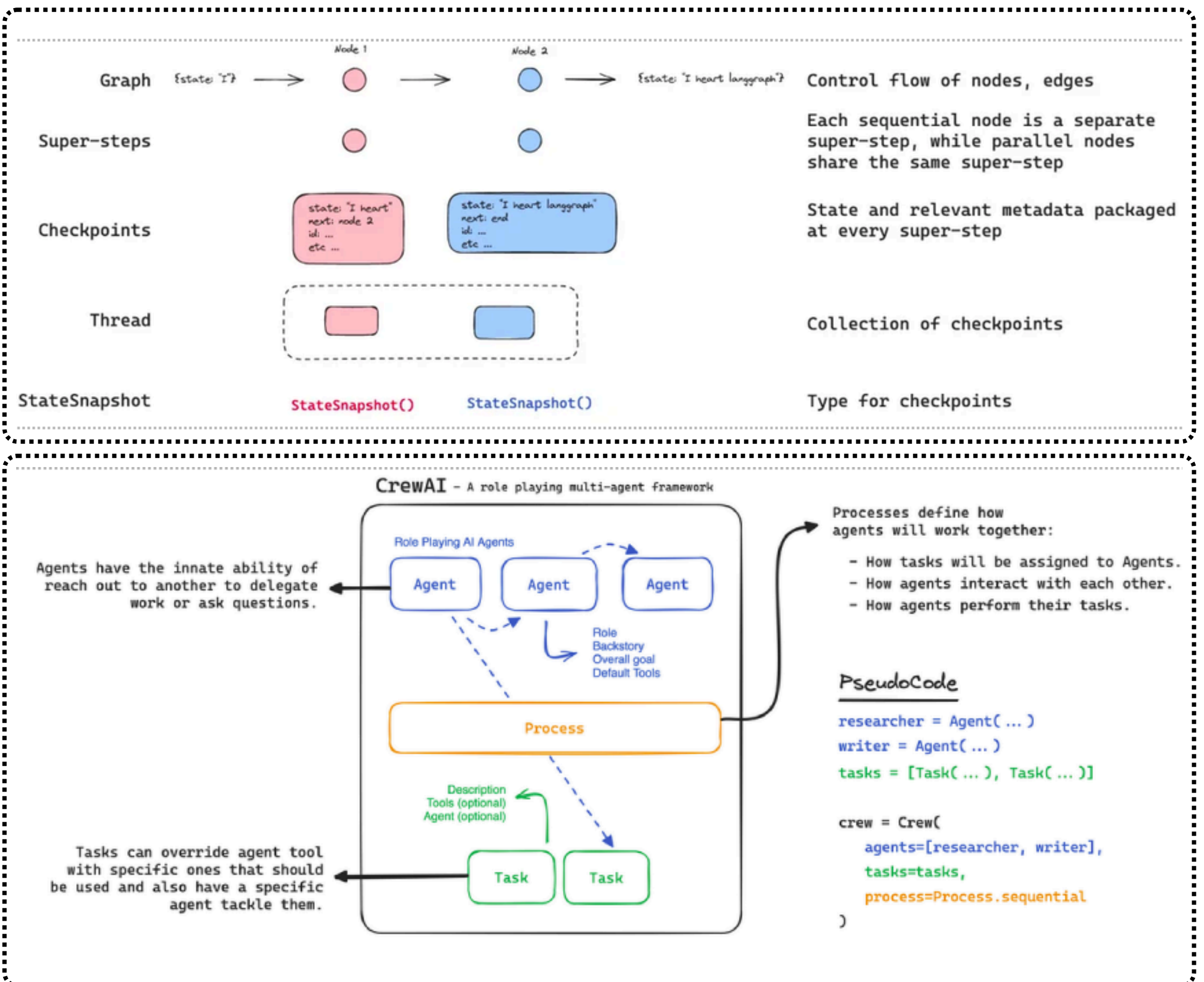


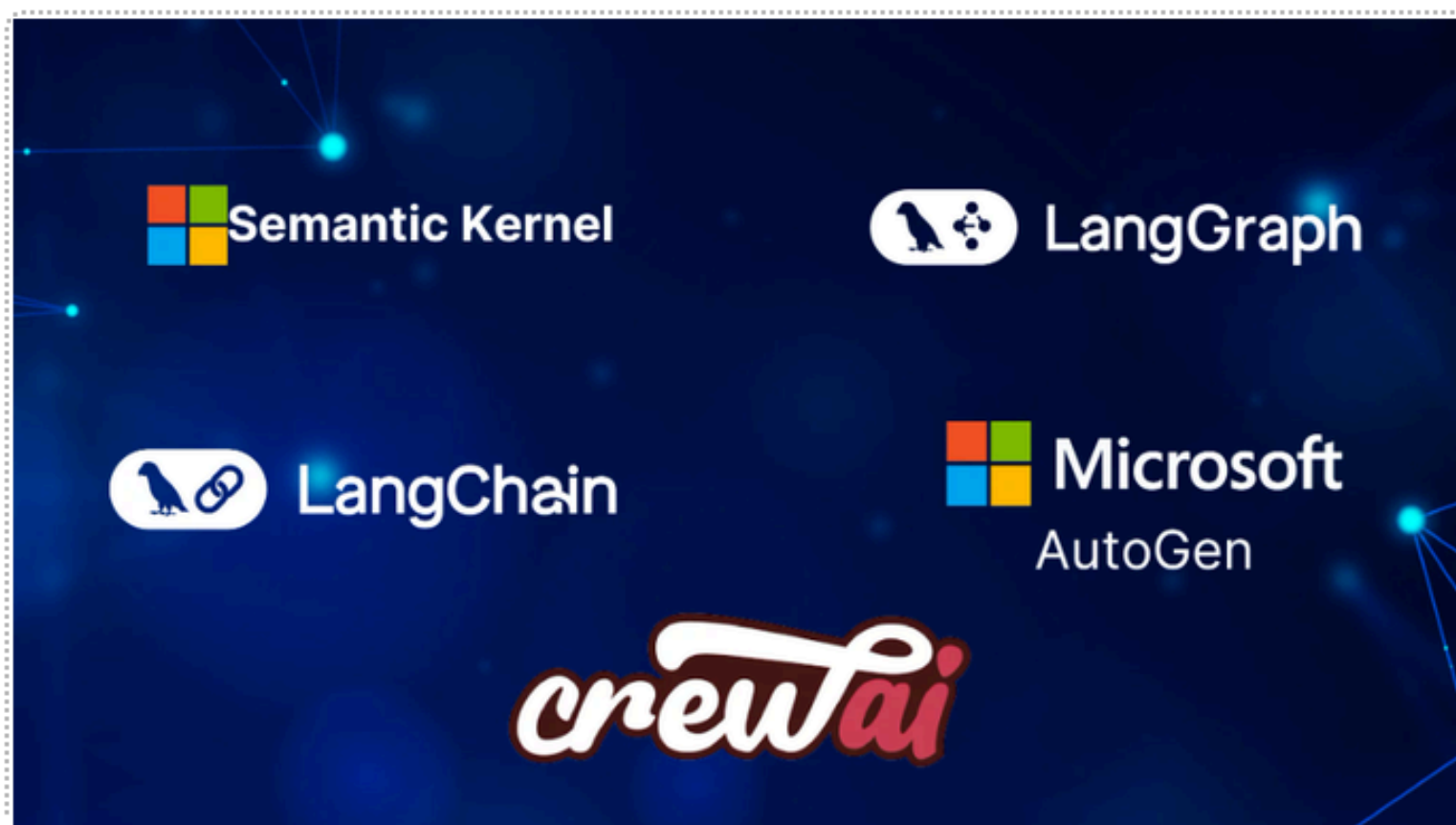
Day 17 of Agentic AI

Frameworks for Agentic AI



#30DaysofAgenticAI

Frameworks for Agentic AI: An Overview



Agentic AI frameworks are essential for orchestrating multi-agent workflows, enhancing efficiency in complex problem-solving. These frameworks enable agents to work autonomously or semi-autonomously, collaborating to achieve defined objectives.

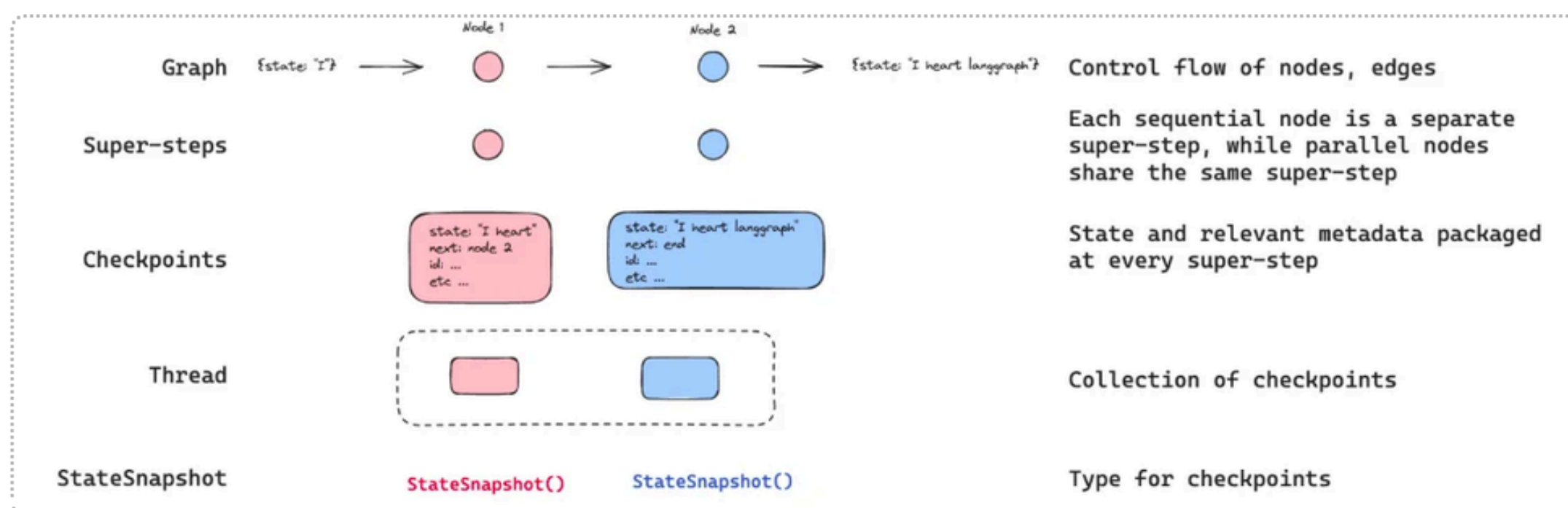
What we'll cover

- **LangGraph:** A graph-based orchestration framework supporting durable execution and human-in-the-loop capabilities.
- **LangChain:** Builds on LangGraph, providing standard model and abstractions a unified agent framework.
- **CrewAI:** Facilitates collaboration among teams of agents for task improved management.
- **AutoGen:** Focuses on conversational multi-agent design and tool integration, enhancing user interaction.



LangGraph: Definition & Functionality

LangGraph is a framework for graph-based orchestration of multi-agent workflows. It enables seamless collaboration among agents, ensuring reliable execution and flexibility.



How it works

- Supports durable execution for ongoing tasks.
- Incorporates checkpointing for workflow reliability.
- Enables human-in-the-loop support for oversight and adjustments.

Key example

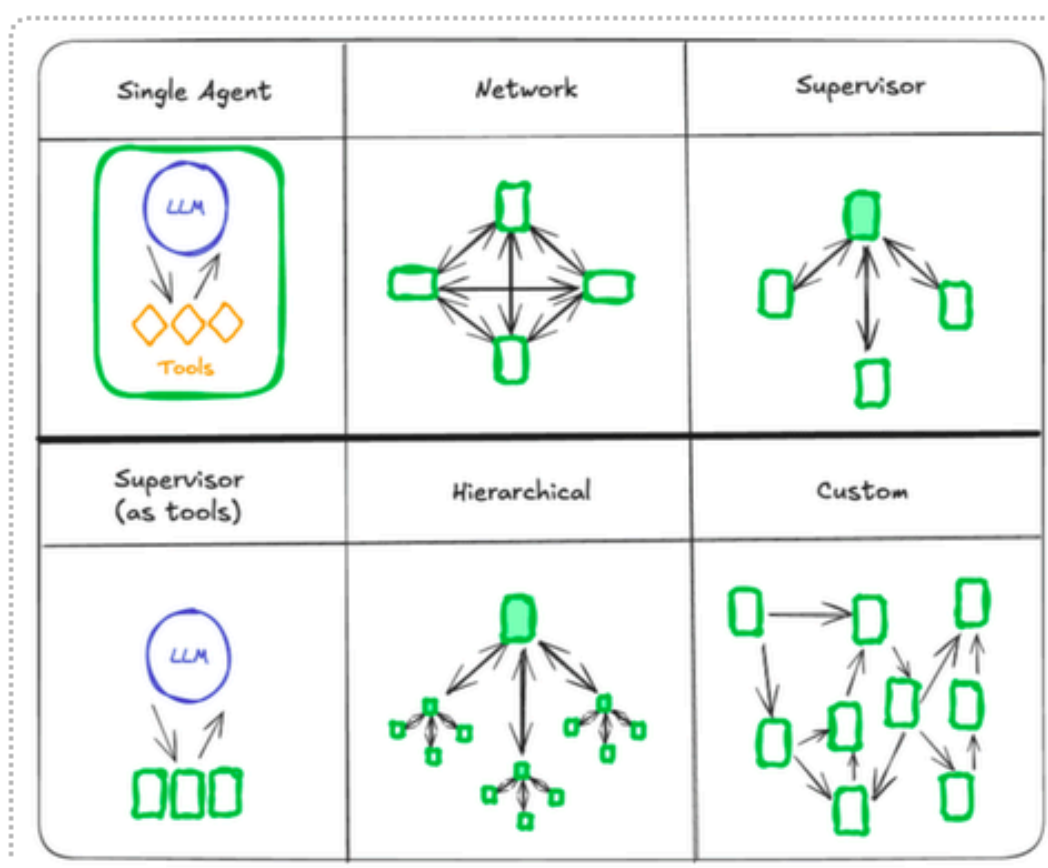
In a logistics operation, LangGraph coordinates multiple agents to manage delivery scheduling and routing effectively.



LangGraph: Key Characteristics

Memory

- Maintains state information across agent interactions.
- Facilitates continuity in multi-agent workflows.
- **Example:** Agents recalling previous tasks to improve efficiency.



Checkpointing

- Allows workflows to save progress at intervals.
- Ensures recovery from failures without starting over.
- **Example:** Resuming a logistics task after a disruption.

Streaming

- Supports real-time data processing and updates.
- Enhances responsiveness of agents during execution.



LangGraph: Use Cases

Use Case 1

Logistics Management

- **Context:** Managing complex delivery operations.
- **Application:** Coordinating agents for scheduling and routing.
- **Benefits:** Improved efficiency in delivery times.

Use Case 2

Workflow Automation

- **Context:** Automating repetitive tasks across agents.
- **Application:** Agents executing tasks based on predefined workflows.
- **Benefits:** Increased productivity and reduced errors.

Use Case 3

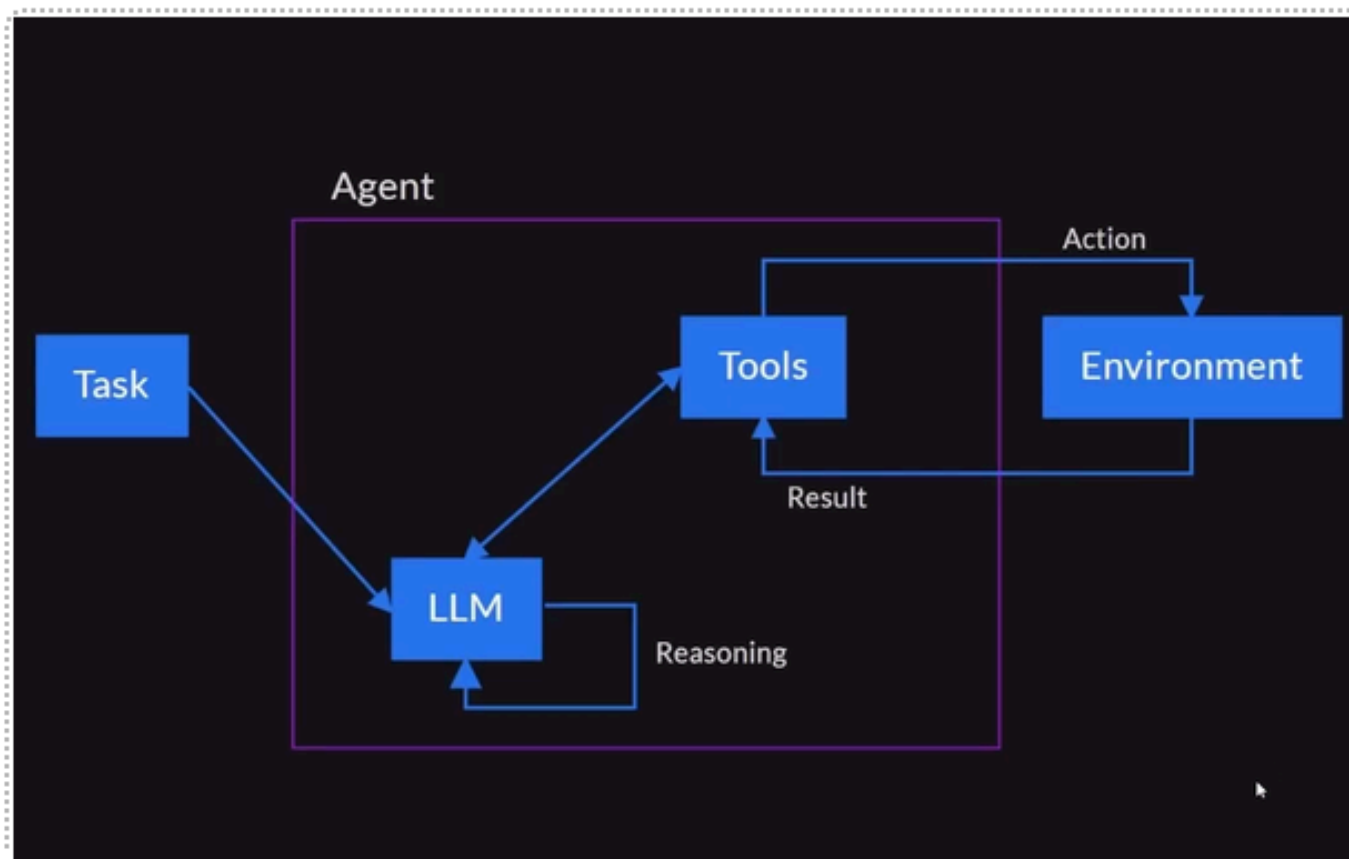
Data Processing

- **Context:** Handling large datasets with multiple agents.
- **Application:** Agents processing data streams for analysis.
- **Benefits:** Faster insights and decision-making.



LangChain: Overview & Updates

LangChain is a framework that builds on LangGraph, enhancing multi-agent workflows with advanced capabilities. It integrates standard model abstractions and a unified agent framework for streamlined development.



Key characteristics

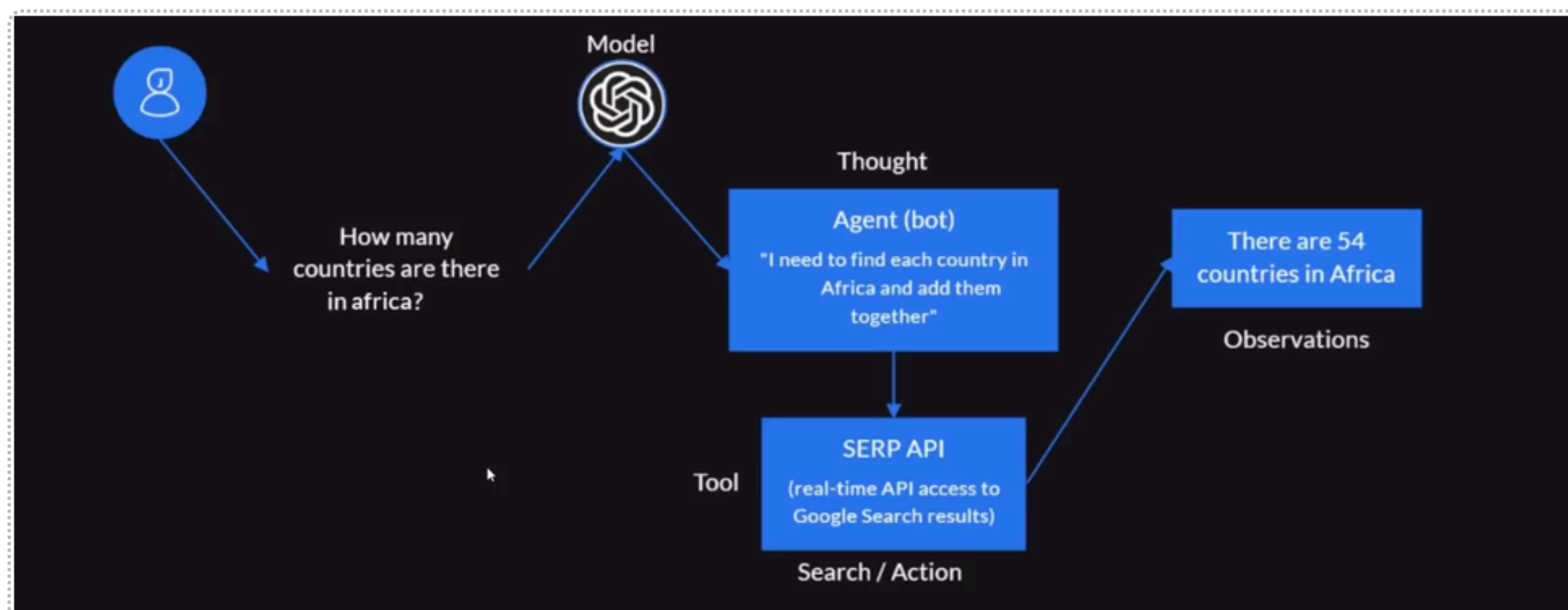
- **Standard model abstractions:** Simplifies integration of various AI models.
- **Unified agent abstraction:** Facilitates extensibility via `create_agent`.
- **Middleware in the agent loop:** Enhances flexibility and adaptability.



LangChain: Characteristics

Structured message content

- Organizes messages for clarity and reasoning.
- Supports citations and tool calls within workflows.
- **Example:** Agents referencing data sources in responses.



Leaner core package

- Streamlined framework for better performance.
- Focuses on essential features for developers.
- **Example:** Faster load times and reduced complexity.

Legacy features migration

- Moves outdated features to a langchain-legacy module.
- Keeps the core package modern and efficient.



LangChain: Applications

Use Case 1

Reasoning-First Agents

- **Context:** Complex problem-solving scenarios.
- **Application:** Agents executing multi-step reasoning tasks.
- **Benefits:** Enhanced decision-making capabilities.

Use Case 2

Advanced Tool Use

- **Context:** Integration of various tools for agents.
- **Application:** Agents utilizing tools for specific tasks.
- **Benefits:** Increased efficiency and task accuracy.

Use Case 3

Multi-Step Problem Solving

- **Context:** Tasks requiring sequential actions.
- **Application:** Agents coordinating to solve intricate problems.
- **Benefits:** Improved outcomes through structured approaches.



CrewAI: Collaborative Agent Teams

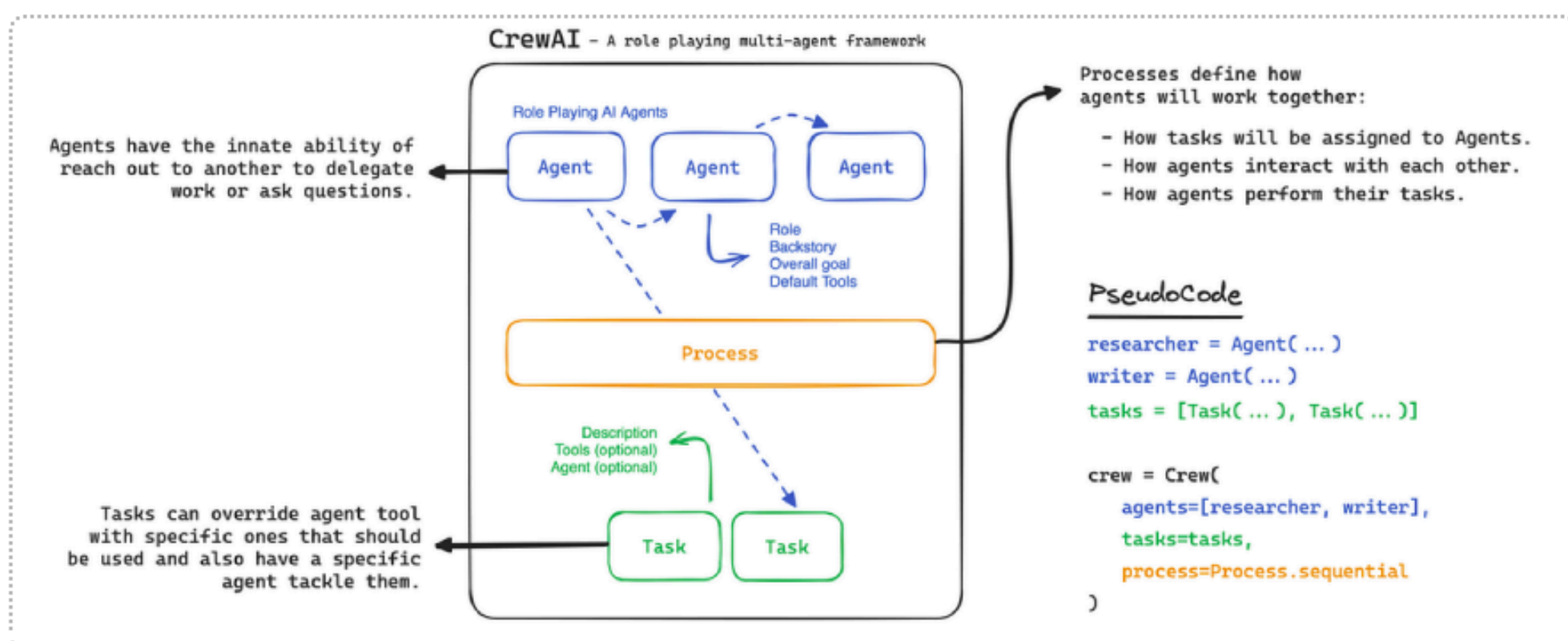
CrewAI is a framework designed for building teams of agents that collaborate effectively on tasks. It enhances agent interaction and coordination for improved outcomes.

Key characteristics

- **Team building:** Facilitates the formation of agent teams based on strengths.
- **Collaboration:** Promotes real-time communication among agents.
- **Agent interaction:** Enables seamless task delegation and management.

Best for

Situations requiring input from multiple agents working together, such as customer support and project management.



AutoGen: Conversational Multi-Agent Design

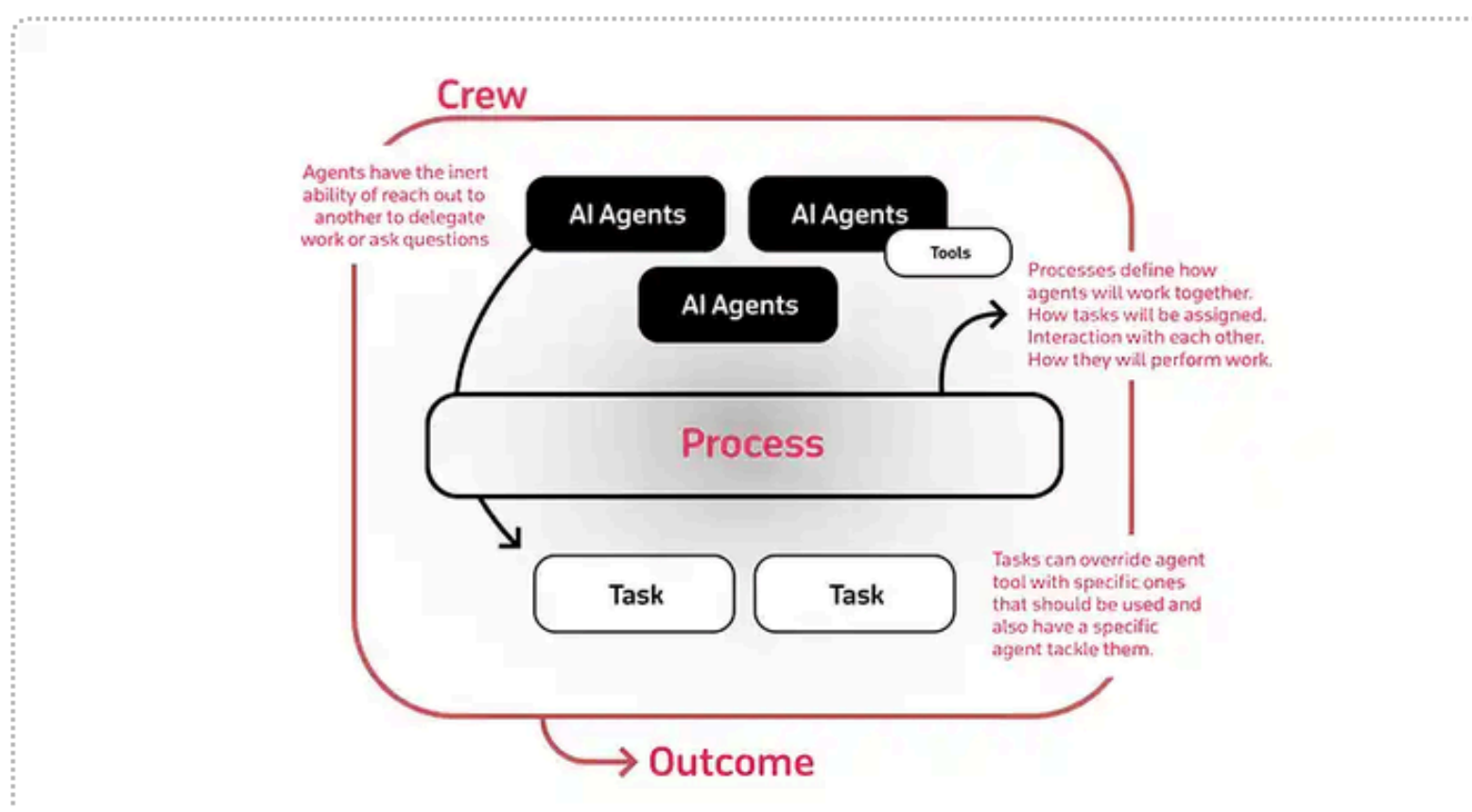
AutoGen is a framework for designing conversational agents that integrate various tools to enhance user interaction. It focuses on creating responsive and capable conversational interfaces.

Key characteristics

- **Conversational design:** Optimizes interactions for user engagement.
- **Tool integration:** Allows agents to access and utilize external resources.
- **Multi-agent systems:** Supports collaboration among multiple conversational agents.

Best for

Developing interactive interfaces, such as virtual assistants that can manage tasks across different platforms.



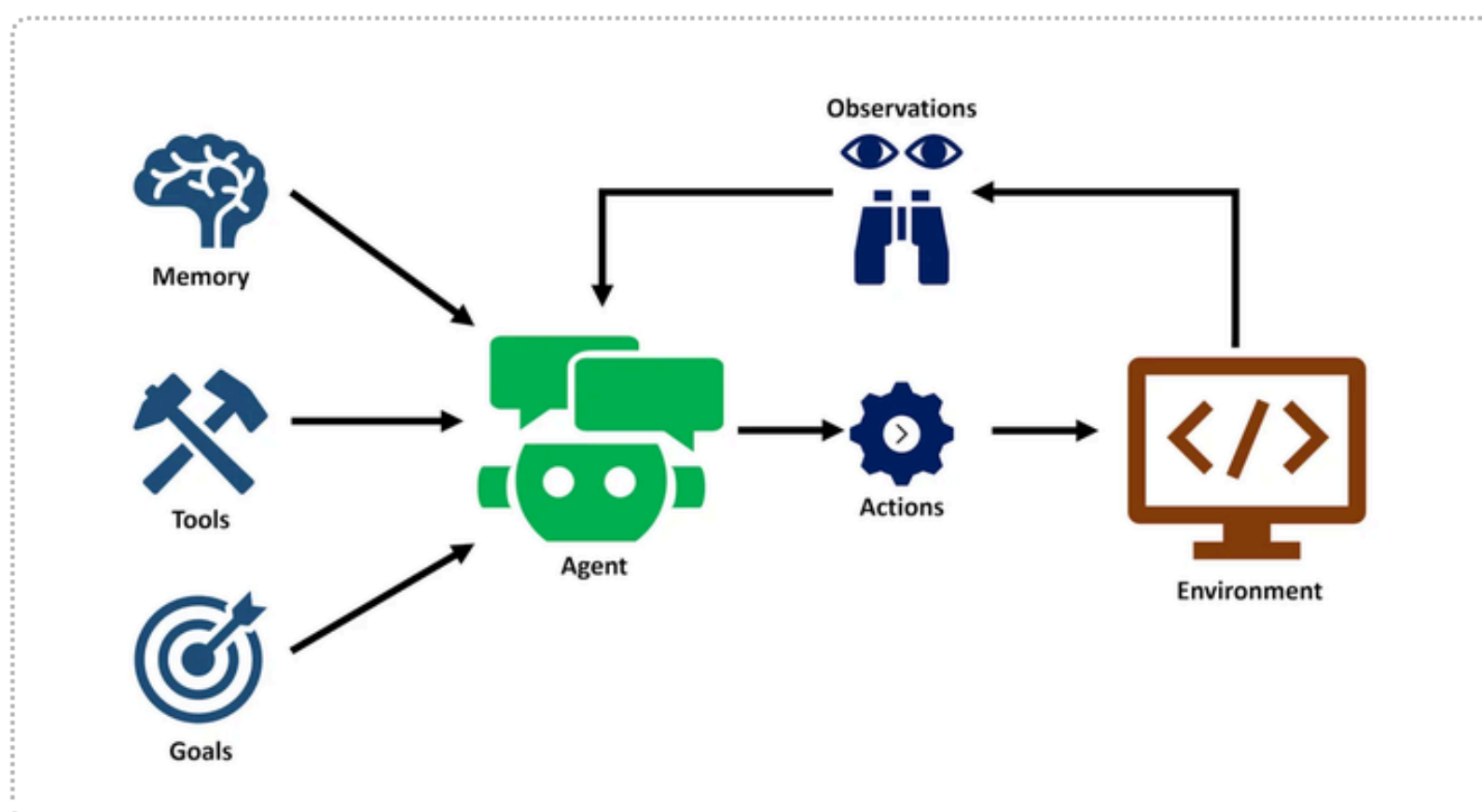
Emerging Frameworks: Future Outlook

Emerging frameworks like OpenAI Swarm and LlamaIndex Agents are set to enhance multi-agent orchestration.

Key features

These frameworks will likely redefine how agents interact and collaborate, leading to more efficient AI systems.

- **Richer orchestration:** Improved coordination among agents for complex tasks.
- **Multi-agent systems:** Enhanced capabilities for collaborative problem-solving.
- **2024+ developments:** Anticipated advancements in agent and technology frameworks.



Comparison of Agentic AI Frameworks

LangGraph

- Focus on graph-based orchestration.
- Supports durable execution and human-in-the-loop.

Best for: Managing complex workflows with multiple agents.

LangChain

- Emphasis on model integration and pipelines.
- Provides structured message handling.

Best for: Applications requiring integration of various LLMs.

CrewAI

- Focus on team collaboration and real-time interaction.
- Enhances task management among agents.

Best for: Tasks requiring input from multiple agents.

AutoGen

- Specializes in conversational design and tool integration.
- Supports multi-agent interactions.

Best for: Creating interactive conversational interfaces.

